

Signed Laplacians for Constrained Graph Clustering

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Motivation & Background



What is clustering?

Problem Definition:

- Group vertices that are close to each other(has many edges between them) into k clusters.
- k is given in the problem.

Standard Clustering:

- Uses only the basic graph structure(vertices and edges) to group vertices

Constrained Clustering:

- Incorporates extra knowledge that are given
- MUST-LINK constraints: vertices that belong in the same cluster(denoted by graph G)
- CANNOT-LINK constraints: vertices that must be separated(denoted by graph H)



Real-world example

Example:

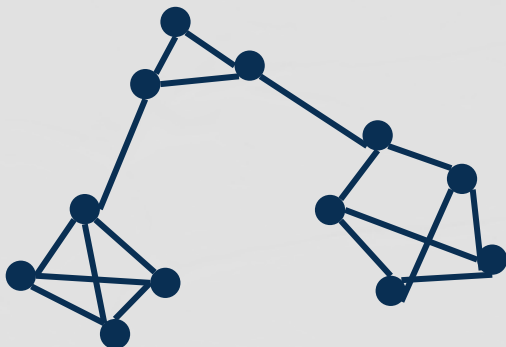
- Grouping weather stations by climate zone. Raw data: geographic distances.
- Stations A and B always record similar temperatures → MUST-LINK
- Stations C and D are in different climate zones → CANNOT-LINK
- Standard clustering ignores this — constrained clustering uses it.



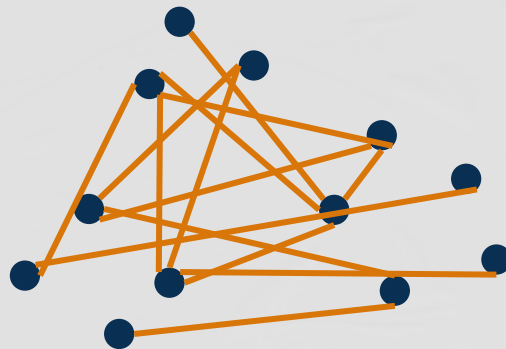
Why not spectral clustering?

Spectral Clustering:

- Uses the second eigenvalue of the Laplacian of the graph
- Looks at only graph G
- Struggles when clusters overlap and need extra information



G(MUST-LINK)



H(CANNOT-LINK)

Background- Spectral Clustering

Spectral Clustering:

- Laplacian $L = D - A$
- Normalized Laplacian $L_{norm} = I - D^{-\frac{1}{2}}AD^{-\frac{1}{2}}$
- We want to minimize the conductance $\phi(S) = \frac{w(S, V \setminus S)}{vol(S)}$
- Dirichlet energy $\sum_{u,v \in E} (f(u) - f(v))^2 = f^T L f$
- Rayleigh quotient property: $\lambda_2 = \min_{f \neq 0, f \perp 1} \frac{f^T L f}{f^T f}$
- Using the second smallest eigenvalue gives a vector with small Dirichlet energy, making it more likely to select a set with low conductance.



Background- Spectral Clustering

Spectral Clustering:

- Cheeger's Inequality ensures that sweeping on the second eigenvector gives a good cut.
- Cheeger's inequality

There is a sweep set S such that $\phi(S) \leq \sqrt{2R(f)}$, for some f , where $R(f)$ is the Rayleigh quotient



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Experimental results

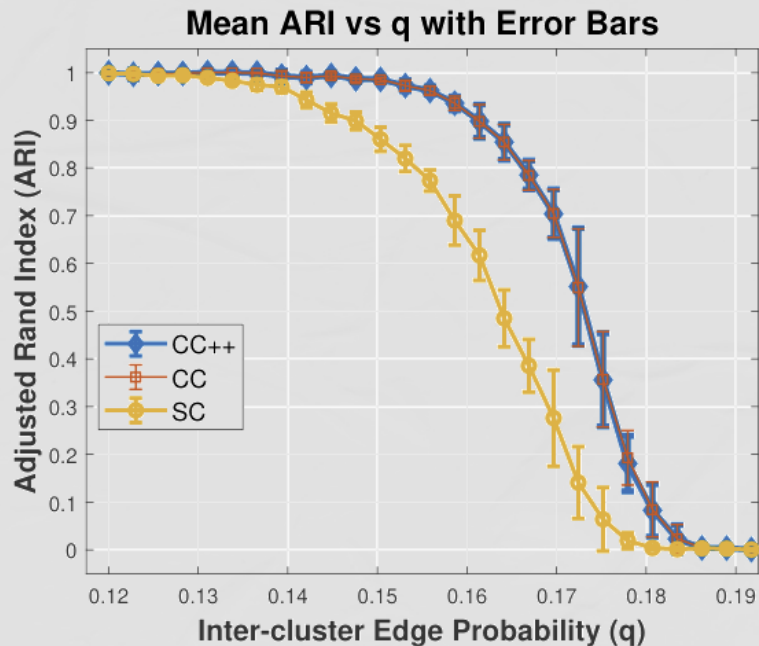


ARI comparison

p: intra-cluster edge probability

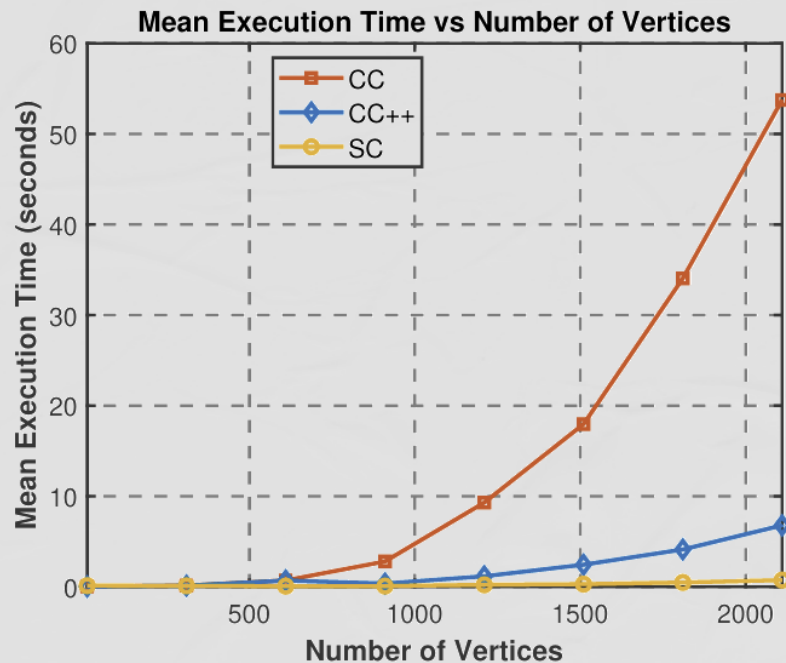
q: inter-cluster edge probability

- Constrained Clustering(CC and CC++) are better than Spectral Clustering(SC) in classifying data according to their known values
- As q increases, distinction between two clusters decrease
- ARI: Adjusted Rand Index. Measures the similarity between two data clusterings or partition



Execution time comparison

- Basic Constrained Clustering(CC) takes much longer than Spectral Clustering(SC).
- The paper's method, Constrained Clustering with loops(CC++) takes much less and CC, being more similar to Spectral Clustering(SC).



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Main algorithm



Main algorithm

1. scale all edge weights so that the $\deg^G(v) = \deg^H(v)$ for all v
2. Compute the normalized Laplacians for G and H
3. Add a small-weighted negative self loop to one vertex in H
4. Solve the generalized eigenvalue problem

$$\frac{\langle f, \Delta^G f \rangle}{\langle f, \Delta^H f \rangle} \text{ subject to } f \perp \mathbb{1}$$

where f is the eigenvector that minimizes the ratio.

5. Apply the sweep-set algorithm on the eigenvector f to partition the vertices of G into two clusters.



Main algorithm

Standard eigenvalue problem

- $L_G x = \lambda x$

Generalized eigenvalue problem

- $L_G x = \lambda L_H x$, where B is symmetric positive definite
- We can convert it to a standard eigenvalue problem if L_H is invertible, by $L_H^{-1} L_G x = \lambda x$
- Usually the Laplacian is singular (not invertible). But adding a negative self-loop makes the matrix invertible
- Basic Constrained Clustering(CC) uses the generalized eigenvalue problem, whereas this paper's algorithm(CC++) uses the standard eigenvalue problem by adding a negative self-loop. The correctness of this method is also proved in the paper.



Main algorithm

Application of the sweep-set algorithm

- The application of the sweep-set algorithm is guaranteed by the Cheeger-type inequality, one of the paper's main results



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Theoretical results



Main result

Cheeger-type inequality for Constrained Clustering

- Let $G = (V, E, w)$ and $H = (V, E', w')$ be graphs such that $\deg^G(v) = \deg^H(v)$ for all $v \in V$.
- Then, it holds that $\phi_H^G \leq 4 \sqrt{\frac{\lambda(\Delta_H^G)}{\lambda_2(\Delta^H)}}$ where Δ^H denotes the normalized Laplacian of the graph H , Δ_H^G is the operator given by $\Delta_H^G(x) = \frac{\langle x, \Delta^G x \rangle}{\langle x, \Delta^H x \rangle}$ and $\lambda_2 = \min_{x \perp 1} \frac{\langle x, \Delta^G x \rangle}{\langle x, \Delta^H x \rangle}$.
- Moreover, the cut achieving this approximation guarantee can be found with a sweep set algorithm.



Main result

Cheeger-type inequality for Constrained Clustering

- We want to prove the existence of a non-empty, proper subset $S \subset V$ such that $\frac{w_G(S, V \setminus S)}{w_H(S, V \setminus S)} =$

$$4 \sqrt{\frac{\langle x, \Delta^G x \rangle \cdot \langle x, x \rangle^2}{x, \Delta^H x}}$$

- To reduce the problem, we find the solution for $z = cx$, where $z_n^w + z_1^2 = 1$.

- Now we want to find a probability density function over $[z_1, z_n]$, and prove $\frac{\mathbb{E}(w_G(S, V \setminus S))}{\mathbb{E}(w_H(S, V \setminus S))} =$

$$4 \sqrt{\frac{\langle z, \Delta^G z \rangle \cdot \langle z, z \rangle^2}{z, \Delta^H z}}$$

- Using the Cauchy-Schwarz inequality, the definition of Δ , and the property $\deg^G(v) = \deg^H(v)$, we can derive the above.
- Define a distribution and choose t according to the probability density function $2|t|$. From this distribution, the bounds of the expectation of the cut ratio matches the intended result.

Main result

Correctness of the idea adding a negative self-loop

- Let $H = (V, E, w)$ be a weighted graph, and let $H' = (V, E', w')$ be another weighted graph such that $E' = E \cup (v_0, v_0)$, where v_0 is a vertex with a self-loop. Assume that $w'|_E = w$ and consider the signature $s = 1$ for all E and $s_{(v_0, v_0)} = -1$. Then, $0 < \lambda_1(\Delta_\alpha^{H'}) \leq \lambda_2(\Delta^H)$, where Δ^H is the Laplacian of graph H .
- Since $\lambda_1(\Delta_\alpha^{H'}) \leq \lambda_2(\Delta^H)$, we can use $\lambda_1(\Delta_\alpha^{H'})$ as the smallest nontrivial eigenvalue



**Thank
you!**

